

Executive Summary

This study guide covers the fundamental principles of biology, spanning cellular structures, metabolic processes like photosynthesis and cellular respiration, and genetics. It also explores cell transport, division, protein synthesis, enzymatic functions, and the cooperative systems that maintain homeostasis in the human body.

KEY TAKEAWAYS

- The cell is the basic structural and functional unit of all living organisms, and all cells arise from pre-existing cells.
- Photosynthesis converts light energy into chemical energy, represented by the summary equation $6\text{CO}_2 + 6\text{H}_2\text{O} = \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$.
- The summary equation of cellular respiration is $\text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2 = 6\text{CO}_2 + 6\text{H}_2\text{O} + \text{ATP}$.
- Cell membranes are selectively permeable barriers that control the movement of substances via passive diffusion, osmosis, and active transport.
- DNA replication, transcription, and translation are the sequential processes by which genetic instructions are copied and expressed as proteins.
- Homeostasis is the maintenance of a stable internal environment, such as keeping human body temperature at approximately 37 degrees Celsius.
- Human organ systems, including the circulatory, respiratory, nervous, and endocrine systems, coordinate to transport nutrients and regulate bodily processes.

Topics

TOPIC 1

Chapter 1: Cellular Biology

The cell is the basic structural and functional unit of all living organisms, as outlined by the fundamental principles of cell theory.

Cell Theory and Organelles

- All living things are composed of cells, and new cells are produced from existing cells.
- The nucleus is the membrane-bound organelle that contains the genetic material.
- The mitochondria is the organelle responsible for producing ATP through cellular respiration.

Chapter 2: Photosynthesis

Photosynthesis is the process by which green plants transform light energy into chemical energy within the chloroplasts.

Light Absorption and Chemical Equations

- Chlorophyll is the green pigment that absorbs light energy.
- The overall equation for photosynthesis shows that carbon dioxide and water, in the presence of light, produce glucose and release oxygen.
- The summary equation of photosynthesis is $6\text{CO}_2 + 6\text{H}_2\text{O} = \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$.

Stages of Photosynthesis

- The light-dependent reactions occur in the thylakoid membranes and produce ATP and NADPH.
- The Calvin cycle is the light-independent stage that fixes carbon dioxide into glucose.

Chapter 3: Cellular Respiration

Cellular respiration consists of three main stages that break down glucose to produce ATP, alongside anaerobic fermentation options.

Stages of Respiration

- Glycolysis occurs in the cytoplasm and is the first stage, during which glucose is broken down into two molecules of pyruvate.
- The Krebs cycle occurs in the mitochondrial matrix.
- The electron transport chain is the final stage and occurs on the inner mitochondrial membrane.

Anaerobic Respiration and Summary Equation

- Fermentation is the anaerobic process that regenerates NAD^+ when oxygen is unavailable.
- The summary equation of cellular respiration is $\text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2 = 6\text{CO}_2 + 6\text{H}_2\text{O} + \text{ATP}$.

Chapter 4: Cell Transport

Cells utilize passive and active transport mechanisms to control the movement of substances across the selectively permeable cell membrane.

Passive and Active Transport Mechanisms

- The cell membrane is the selectively permeable barrier that controls what enters and leaves the cell.
- Diffusion is the passive movement of molecules from an area of high concentration to low concentration.
- Osmosis is defined as the diffusion of water across a selectively permeable membrane.
- Active transport is the movement of substances against their concentration gradient, requiring energy in the form of ATP.
- Endocytosis is the process by which the cell engulfs large particles by wrapping the membrane around them.

Chapter 5: DNA and Protein Synthesis

DNA carries genetic instructions that are replicated and expressed as proteins through transcription and translation.

DNA Replication and Transcription

- DNA replication is the process by which a double-stranded DNA molecule is copied to produce two identical DNA molecules.
- Transcription is the process by which the DNA sequence of a gene is copied into messenger RNA.

Translation and the Genetic Code

- Translation is defined as the process by which the ribosome builds a protein from the mRNA sequence.
- Each three-nucleotide codon on mRNA specifies one amino acid.
- The genetic code is the set of rules by which information encoded in genetic material is translated into proteins.

Chapter 6: Cell Division

The cell cycle consists of interphase, mitosis, and cytokinesis to facilitate growth and nuclear division.

The Cell Cycle and Mitosis

- Mitosis is the process of nuclear division that produces two genetically identical daughter cells.
- The cell cycle consists of interphase, mitosis, and cytokinesis.
- Interphase is the stage during which the cell grows, copies its DNA, and prepares for division.

Stages of Mitosis

- Prophase is the first stage of mitosis, during which chromosomes condense.
- Anaphase is the stage during which sister chromatids are pulled apart to opposite poles.

Chapter 7: Genetics

Genetics studies heredity through genes and alleles, determining an organism's genotype and observable phenotype.

Genes, Alleles, and Expression

- A gene is defined as the basic unit of heredity, a segment of DNA that codes for a specific trait.
- An allele is an alternative form of a gene at the same locus.
- The genotype is the genetic makeup of an organism, while the phenotype is its observable characteristics.
- A dominant allele is expressed when only one copy is present, whereas a recessive allele is only expressed when two copies are present.
- Punnett squares are used to predict the probability of offspring inheriting particular traits.

Chapter 8: Enzymes

Enzymes are protein catalysts that speed up chemical reactions and are highly sensitive to environmental conditions.

Enzyme Action and Models

- An enzyme is defined as a protein that acts as a biological catalyst, speeding up chemical reactions without being consumed.
- The active site is the region of an enzyme where the substrate binds.
- The induced fit model describes how the enzyme changes shape to accommodate the substrate.

Temperature and Denaturation

- Optimal temperature is the temperature at which an enzyme works fastest.
- Denaturation is defined as the loss of an enzyme's three-dimensional shape, which destroys its function.

Chapter 9: Homeostasis

Homeostasis maintains a stable internal environment using feedback mechanisms and hormonal regulation.

Feedback Systems and Temperature

- Negative feedback is a mechanism that reverses a change, bringing the system back to its set point.
- Positive feedback amplifies a change and drives the system further from its set point.
- The normal human body temperature is maintained at approximately 37 degrees Celsius.

Hormonal Regulation of Glucose

- Blood glucose concentration is regulated by the hormones insulin and glucagon.
- Insulin is secreted when blood glucose is high, promoting glucose uptake.

Chapter 10: Human Systems

Human organ systems coordinate rapid and long-term processes to transport nutrients, exchange gases, and maintain life.

Transport and Gas Exchange

- The circulatory system transports oxygen, nutrients, and waste products throughout the body, powered by the muscular heart.
- The respiratory system is responsible for gas exchange, bringing oxygen into the body and removing carbon dioxide.

Coordination and Regulation

- The nervous system coordinates rapid responses through electrical signals called nerve impulses.
- The endocrine system regulates long-term processes through chemical messengers called hormones.

Terms & Definitions

Term	Definition
Cell	The basic structural and functional unit of all living organisms.
Cell theory	A fundamental principle of biology that states that all living things are composed of cells and that new cells are produced from existing cells.
Nucleus	The membrane-bound organelle that contains the genetic material.
Mitochondria	The organelle responsible for producing ATP through cellular respiration.
ATP	A molecule produced by mitochondria through cellular respiration, providing energy for cellular processes.
Photosynthesis	The process by which green plants transform light energy into chemical energy.
Chloroplast	The organelle where photosynthesis occurs, containing the green pigment chlorophyll.
Chlorophyll	The pigment that absorbs light energy during photosynthesis.
Glucose	A sugar produced by photosynthesis, represented by the chemical formula $C_6H_{12}O_6$.
Carbon dioxide	A gas used in photosynthesis and produced in cellular respiration.
Oxygen	A gas released during photosynthesis and used in cellular respiration.
Light-dependent reactions	The reactions in photosynthesis that occur in the thylakoid membranes and produce ATP and NADPH.
NADPH	A molecule produced during the light-dependent reactions of photosynthesis.
Calvin cycle	The light-independent stage of photosynthesis that fixes carbon dioxide into glucose.
Cellular respiration	The process that produces ATP by breaking down glucose in the presence of oxygen.
Glycolysis	The first stage of cellular respiration, occurring in the cytoplasm, where glucose is broken down into pyruvate.
Krebs cycle	The stage of cellular respiration occurring in the mitochondrial matrix.
Electron transport chain	The final stage of cellular respiration occurring on the inner mitochondrial membrane.
Fermentation	The anaerobic process that regenerates NAD^+ when oxygen is unavailable, producing ATP without oxygen.
NAD^+	A molecule regenerated during fermentation when oxygen is unavailable.
Cell membrane	The selectively permeable barrier that controls what enters and leaves the cell.

Term	Definition
Diffusion	The passive movement of molecules from an area of high concentration to low concentration.
Osmosis	The diffusion of water across a selectively permeable membrane.
Active transport	The movement of substances against their concentration gradient, requiring energy in the form of ATP.
Endocytosis	The process by which the cell engulfs large particles by wrapping the membrane around them.
DNA	The molecule that carries the genetic instructions for the development and functioning of living organisms.
DNA replication	The process by which a double-stranded DNA molecule is copied to produce two identical DNA molecules.
Transcription	The process by which the DNA sequence of a gene is copied into messenger RNA.
mRNA	Messenger RNA that carries the genetic code from DNA to the ribosome for protein synthesis.
Translation	The process by which the ribosome builds a protein from the mRNA sequence.
Codon	A three-nucleotide sequence on mRNA that specifies one amino acid.
Genetic code	The set of rules by which information encoded in genetic material is translated into proteins.
Mitosis	The process of nuclear division that produces two genetically identical daughter cells.
Cell cycle	The cycle consisting of interphase, mitosis, and cytokinesis.
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Prophase	The first stage of mitosis, during which chromosomes condense.
Anaphase	The stage during which sister chromatids are pulled apart to opposite poles of the cell.
Gene	The basic unit of heredity, a segment of DNA that codes for a specific trait.
Allele	An alternative form of a gene at the same locus.
Genotype	The genetic makeup of an organism.
Phenotype	The observable characteristics of an organism.
Dominant allele	An allele expressed when only one copy is present.

Term	Definition
Recessive allele	An allele expressed only when two copies are present.
Punnett square	A tool used to predict the probability of offspring inheriting particular traits.
Enzyme	A protein that acts as a biological catalyst, speeding up chemical reactions without being consumed.
Active site	The region of an enzyme where the substrate binds.
Substrate	The molecule upon which an enzyme acts.
Induced fit model	A model describing how an enzyme changes shape to accommodate the substrate.
Optimal temperature	The temperature at which an enzyme works fastest.
Denaturation	The loss of an enzyme's three-dimensional shape, which destroys its function.
Homeostasis	The maintenance of a stable internal environment despite external changes.
Negative feedback	A mechanism that reverses a change, bringing the system back to its set point.
Positive feedback	A mechanism that amplifies a change and drives the system further from its set point.
Circulatory system	The system that transports oxygen, nutrients, and waste products throughout the body.
Heart	The muscular organ that pumps blood through the circulatory system.
Respiratory system	The system responsible for gas exchange, bringing oxygen into the body and removing carbon dioxide.
Nervous system	The system that coordinates rapid responses through electrical signals called nerve impulses.
Endocrine system	The system that regulates long-term processes through chemical messengers called hormones.
Insulin	A hormone secreted when blood glucose is high, promoting glucose uptake.
Glucagon	A hormone that regulates blood glucose concentration.
Summary equation of photosynthesis	$6\text{CO}_2 + 6\text{H}_2\text{O} = \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$
Summary equation of cellular respiration	$\text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2 = 6\text{CO}_2 + 6\text{H}_2\text{O} + \text{ATP}$

Key Facts & Formulas

Formula / Fact	Context
respiration is $C_6H_{12}O_6 + 6O_2 = 6CO_2 + 6H_2O + ATP$.	respiration is $C_6H_{12}O_6 + 6O_2 = 6CO_2 + 6H_2O + ATP$.
$C_6H_{12}O_6$	produce glucose and release oxygen. The summary equation of photosynthesis is $6CO_2 + 6H_2O = C_6H_{12}O_6 + 6O_2$. The light-dependent reactions occur in the thylakoid membranes and produce ATP and